



THE PRICE OF THE PEAK



A report by
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COM-480, Data Visualization

28 May 2026



Introduction

Mountains have always carried a certain weight beyond the physical climb. For over a century, the Himalayas have drawn explorers, national expeditions, record-chasers, and eventually paying tourists from every part of the world. The story of how that happened, how Everest in particular shifted from a remote and almost mythological objective into something organized, commercialized, and accessible to people without elite mountaineering backgrounds, is one worth telling properly, and it has real data behind it.

Our project is an interactive scrollytelling visual essay that follows that transformation across four chapters: the Himalayas as a geographic and historical backdrop, the early conquest of Everest and its major routes, the commercial turn that changed who goes and under what conditions, and finally the human cost embedded in decades of high-altitude death records. We wanted the visualisations to feel like a natural part of the narrative, not something bolted on to make the page look more technical.

The intended audience is anyone with a passing interest in Everest or mountaineering history, no prior knowledge needed. Readers who already know the story will find something concrete to dig into in the data; readers coming in fresh should be able to follow it from beginning to end without losing the thread. The site is built with D3.js, Three.js, and a scrollytelling architecture.

Dataset & Preprocessing

The project is built on the Himalayan Database, which had the scale and historical detail the project needed: expeditions, climbers, peaks, routes, summit outcomes, deaths, hired staff, oxygen use, trekking agencies, and dates. The central idea from the start was that Everest had shifted from a remote expedition goal into something far more organised and commercial, and this dataset looked strong enough to support that story.

The first technical work was basic EDA. We looked at expeditions over time, popular peaks, seasons, summit success rates, death records, death rates, oxygen use, agency involvement, hired staff numbers, and Everest-specific records. The charts at this stage were simple, mostly bars and lines, the point was just to understand what the data could actually support before committing to a direction.

A few things came out of that exploration clearly: expeditions grew strongly over time, trekking agencies rose from almost nothing to handling nearly all expeditions, death rates moved differently from expedition counts, and hired staff and clients showed different risk patterns. Everest also had enough data to work as the central case study. That gave the project its first real structure, the Himalayas as context, Everest as the main thread, commercialisation as the turning point, and deaths by altitude as the cost.

It is worth mentioning that since the raw data was too large and detailed for the browser, we added a preprocessing step in Python to aggregate it into smaller JSON files, one per chart.

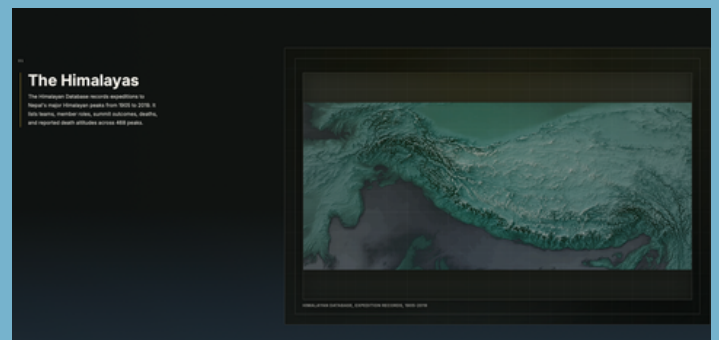
Visualizations

Opening & The Himalayas

The original opening was a simple dark screen with the title centered on it, minimal but unfinished. The Himalayas section that followed used a plain light layout with a photograph and a text card beside it, which worked as a placeholder but didn't give the section much visual weight.

For the final version, the opening was rebuilt around a 3D model of Everest rendered with Three.js, with an LED-style dot post-processing effect applied over it. We tested a heightmap-based approach first, but the GLB model gave a cleaner result. The goal was to make the opening feel visually striking without turning it into a full 3D explorer, so rotation was kept, but zoom and pan were disabled.

The Himalayas section was redesigned into an atlas-style plate with a shaded-relief map. We deliberately avoided adding interaction here, the section's only job is to introduce the geographic scope of the dataset, and a static map treatment does that more cleanly than a live interactive map would.

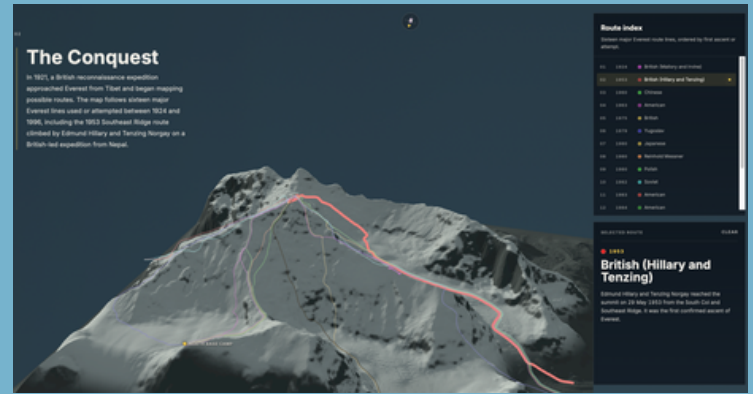


The Conquest

The Conquest chapter went through the most significant visual transformation between milestones. The original version, shown in the left image, placed the mountain as a background element, two static aerial views side by side with route lines drawn over them, and a text panel to the right. It worked as an illustration, but the mountain itself felt secondary to the interface around it.

For the final version, we rebuilt the section entirely around a 3D Everest terrain scene rendered with Three.js. The mountain became the main object, and the route index, the selected-route panel, and the narrative text were organised around it. Sixteen historical route lines are drawn directly onto the terrain, ordered chronologically from 1924 to 1996, and the user can select any of them to read its story. We also added a layer of volumetric clouds drifting around the mountain, shaped so they thin out near the summit and fade at the edges of the scene rather than sitting in front of the routes below — atmosphere, not an obstacle. The shift from a flat image treatment to an interactive, weather-aware 3D scene made a real difference to how the section feels: rather than reading about the routes, you can see them in relation to each other on the actual geometry of the mountain.

The route index panel on the right keeps all sixteen routes visible at once, so the reader can scan the full chronology before committing to anything. Selecting a route brings up a panel with the historical context — who climbed it, when, and what happened — and drops a small bright dot that climbs the chosen line from base to summit on a loop, marking which route you're tracking.

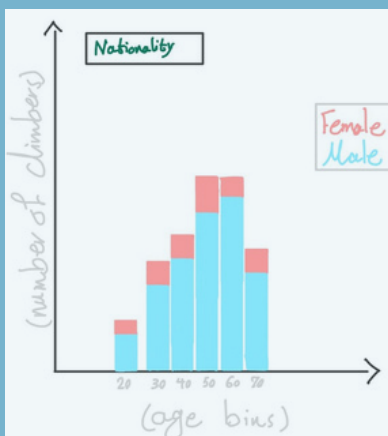


Worth mentioning is that although both the opening section and The Conquest are built around a 3D rendering of Everest, they serve completely different purposes and were built separately to reflect that. The opening model is purely atmospheric, it just needs to look striking and respond to rotation. The Conquest scene had a different job entirely: the geometry needed to be accurate enough for route lines to sit correctly on the terrain, routes needed to be clickable, mountain faces needed labels, and the camera had to be controlled enough that the reader could actually read what they were looking at. One model trying to do all of that while also serving as a visual introduction would have ended up doing neither well.

A Dropped Visualization — The Climbers

One visualization we built but decided to leave out of the final site was a demographic overview of the 73,022 climbers in the dataset; their age distribution, gender ratio, and nationality. The original sketch shows a simple stacked bar chart by age, gender, and nationality filters; the implemented version grew beyond that, adding decade toggles, and summary statistics at the top.

During development we realised the section didn't sit comfortably in the narrative. It showed a lot about climber demographics but didn't carry a clear message on its own, it felt more like an exploratory tool than a chapter in a story. Since the rest of the site is built around a specific arc, adding a section that didn't move that arc forward would have made the whole thing feel less focused. So we made the decision to leave it out.



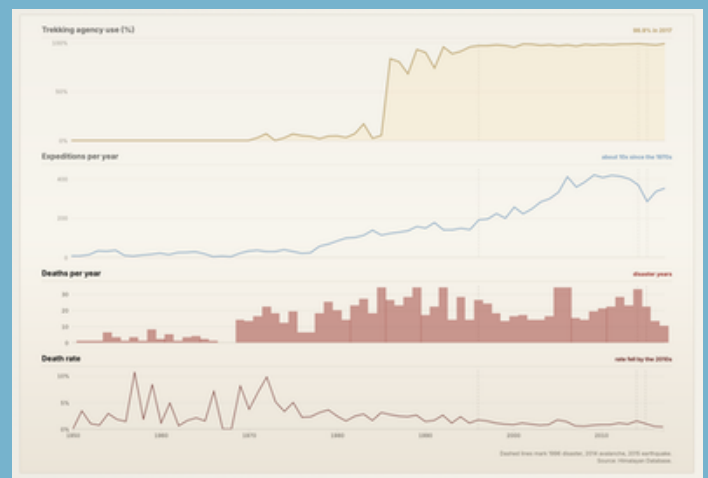
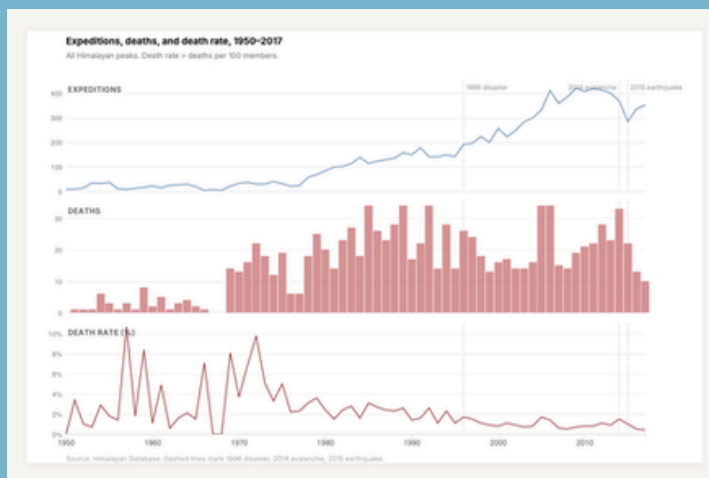
The Turn

The Turn section covers the commercialization of Himalayan climbing, the moment expeditions shifted from rare feats of exploration to a managed industry. The section tells this story through four related measures: trekking agency use, expeditions per year, deaths per year, and death rate.

In the previous version, all three time series (expeditions, deaths, death rate) were stacked into a single SVG panel. While compact, this made it harder to give each metric its own weight. The redesign separates them into four individual charts, adding trekking agency use as the lead chart. This new first panel earns its place, agency use jumping from near 0% in the 1970s to 98.9% by 2017 is the clearest single signal of commercialization, and opening with it frames everything that follows.

The main challenge in this section was presenting four related charts without the layout feeling like a data dashboard. To keep it readable, we moved titles and annotations out of the SVG wherever possible, when everything lives inside the chart area, labels quickly start competing with the data. Now the SVG handles axes and marks, and the HTML around it handles titles and explanatory notes.

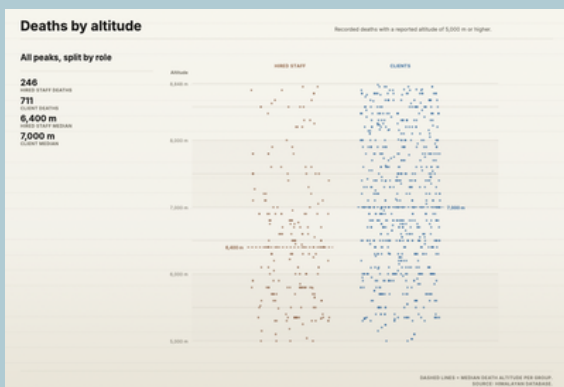
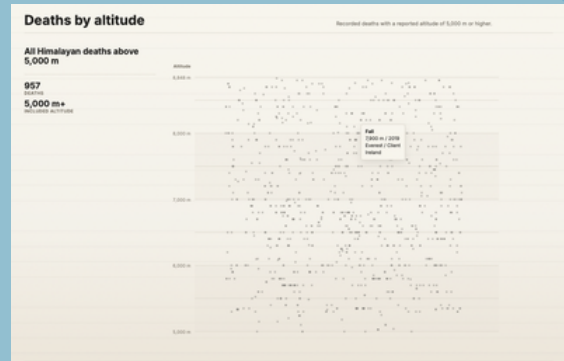
To help readers grasp the key takeaways without reading off the axes, we added endpoint annotations to each chart, callouts like "98.9% in 2017" and "about 10x since the 1970s" placed at the edge of the lines. These act as conclusions built into the chart rather than left to the reader to extract. For the deaths panel, we used a "disaster years" label alongside dashed vertical lines to signal that the visible spikes are named events, not random noise; the specific events are noted in the footnote below the charts rather than cluttering the chart area. Finally, hover tooltips were added so readers can inspect individual year values interactively, keeping the static view clean while still making the underlying numbers accessible.



The Cost

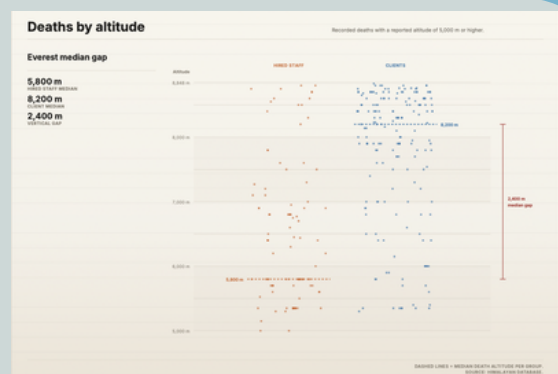
The Cost section visualizes 957 recorded deaths above 5,000 meters as a dot plot, where each dot represents one person placed at the altitude where they died. The section builds through four scrollytelling states: all deaths, deaths split by role, Everest only, and the median altitude gap between hired staff and clients on Everest.

The one dot per person encoding was a deliberate choice. Aggregate charts would have made the deaths feel statistical; placing each one at the altitude where it occurred gives the section its weight without editorializing. The horizontal spread is just to prevent dots from overlapping, only the vertical position, the altitude, carries meaning.



The role split in the second state introduces color, orange for hired staff, blue for clients, and dashed median lines to make the difference legible. Across all peaks, hired staff die at a median of 6,400 meters, clients at 7,000 meters.

Narrowing to Everest in the third state sharpens the contrast: 5,800 meters versus 8,200 meters. The fourth state makes that gap explicit with a bracket annotation on the right edge of the chart, labeling the 2,400-meter vertical difference directly.



Visualization Decision: We considered a mountain silhouette in the background. The idea was to make the altitude axis feel more grounded. But with nearly a thousand dots already on the page, adding a shape behind them just created clutter. The data was doing the job on its own.

The Business

The Business is a new section added for this milestone, slotted in after The Cost as the closing chapter. Its argument is that commercialization didn't just change Everest's economics — it changed who shows up on the mountain. To make that argument we needed four time series read together, so the layout is four small-multiple panels stacked on a shared year axis from 1970 to 2017, with coloured direct-labels at the endpoint of each line and dashed verticals on the 1996 disaster, the 2014 avalanche and the 2015 earthquake.

The panels are revealed in two scroll-driven pairs. The first pair is about the client: the average paying climber aged from 31 to 43, and supplemental oxygen — once the exception — was used by roughly 39 percent of them by the end of the window. The second pair is about the support and the payoff: hired staff grew from 10 to 35 percent of people on the mountain, and the share of clients reaching a summit climbed from 16 to 53 percent. The four series were chosen so they read as one sentence — the client got older and better-equipped, the operation around them got bigger, and the success rate followed.

Worth mentioning is that all four lines move in the same direction once the agencies arrive, and that is the section's argument rather than an accident: an older, oxygen-supported client only becomes viable when the staffing scales to support them, and the success rate is what makes the whole arrangement sellable. The chart is laid out so those four motions land in a single eye-movement rather than four separate readings, and the panels are scoped to paying clients specifically — hired staff are excluded from the age and oxygen series — so what's being measured is the commercial slice of the mountain, not the mountain as a whole.



What We Would Improve

Several directions could extend the project further. The most natural next step would be expanding the 3D treatment beyond Everest to other major Himalayan peaks, allowing users to explore expedition routes across the full range rather than a single mountain. The transition between the Himalayas atlas and The Conquest could also be reworked; a more native visual handoff between the two sections would make the geographic context feel less like a separate plate and more like a continuous journey into the mountain. Finally, the site is not currently optimised for mobile, and given the scrollytelling architecture and the 3D scenes, making it work well on smaller screens would require a meaningful redesign of several sections rather than simple resizing.



Contributions



What We Learned

The biggest thing this project taught us was the difference between having data and having a story. We started with a dataset and a general interest in Everest, it took several rounds of exploration before we understood what the data was actually saying and how to structure it into something a reader could follow. For some of us, this project was the first real opportunity to work with JavaScript, D3, and the broader web stack. But the bigger lesson wasn't technical. Most of the real work happened in the annotations, the panel text, the handoffs between sections, and the decisions about what not to show, and that, more than anything, is what made the project feel like a real data story rather than a collection of charts.

Vsevolod



Worked on:

- Dataset selection and EDA
- Data cleaning and preprocessing
- First scrollytelling prototype for M2
- Final site design and implementation
- Everest figures: research, terrain data, route annotation;
- D3 charts for "The Turn" and "The Cost"
- Sources and Inspiration page
- Screencast recording
- Project timeline for process book

Shahrzad



Worked on:

- Designed and implemented the climber demographic visualisation, which was later removed from the final site
- Writing and content of the Milestone 2 report
- Process book: authored the majority of the written content, visual design, and layout
- Reviewing and editing EDA

Maksim



Worked on:

- Volumetric clouds and better edges in 3D figure
- Climbing dot on the selected route
- Clickable compass to recenter the 3D camera
- Hover tooltips with values on "The Turn" charts
- New "The Business" section
- Web-site loading optimization
- Process book — reviewing, "The Business" section implementation
- Screencast recording



[Github repo](#)



[Website Link](#)